



SCION

Scalable Intelligent Governance Platform

Strategic Initiative for Teradata DNA – Kalido Replacement Proposal

EXECUTIVE & TECHNICAL BRIEFING | INTERNAL CONFIDENTIAL

Executive Summary: Strategic Pivot to Internal IP



STRATEGIC SHIFT

Replaces Kalido within Teradata DNA with proprietary IP. Transitions from vendor-dependency to a modular, internal asset.



FINANCIAL GAIN

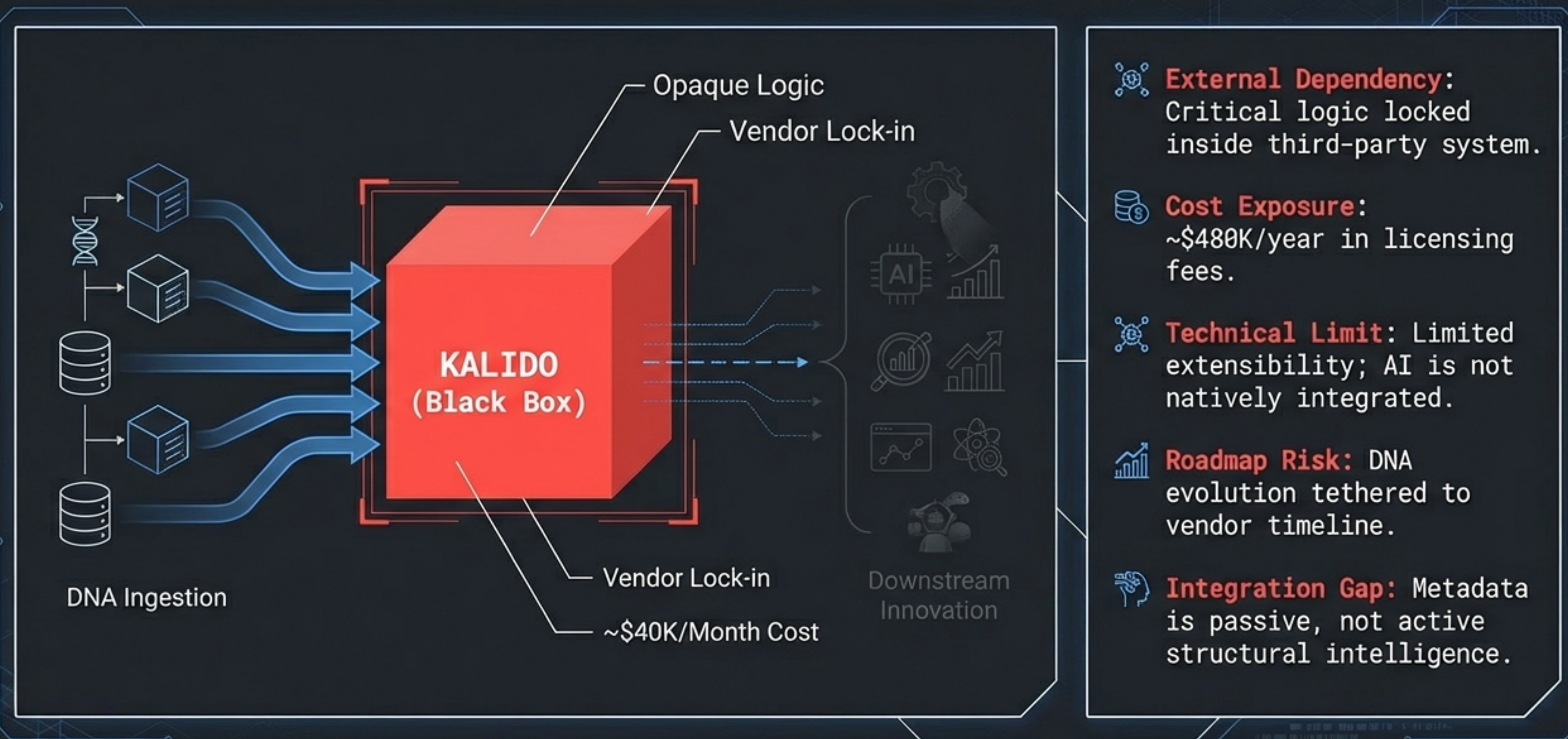
Eliminates ~\$480K annual recurring licensing exposure. Immediate ROI upon decommissioning.



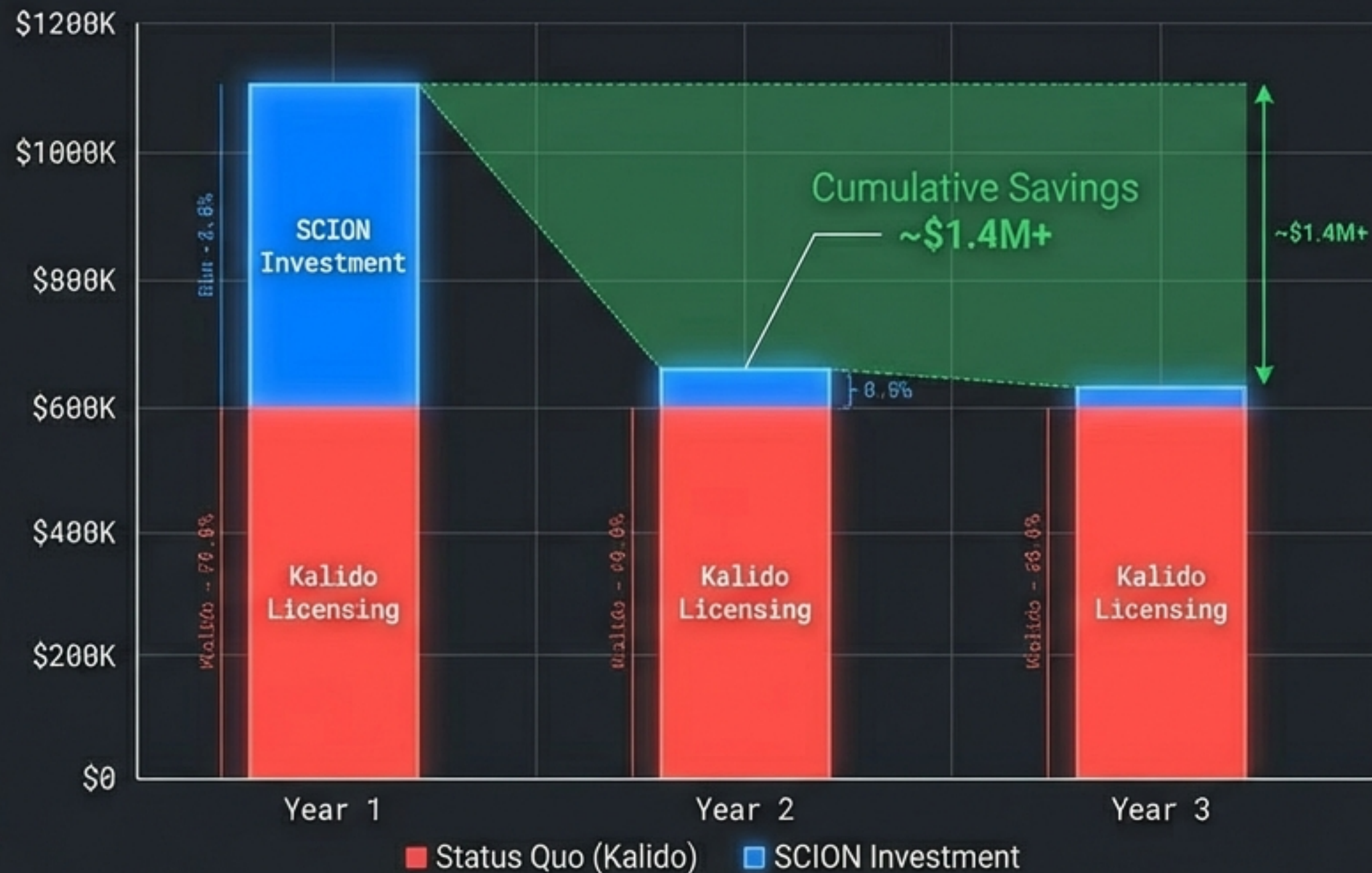
EXECUTION & TECH

5–6 month roadmap to 'DNA-Ready' status. Shifts from black-box logic to deterministic governance + AI.

Strategic Context: The Cost of Black-Box Dependency



Financial Impact & ROI Projection

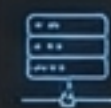


~\$480K

/ Year Savings (Recurring)



OpEx → CapEx
Reallocation



Zero
Marginal Scale Cost



100%
IP Ownership

Vision: From Passive Metadata to Structural Intelligence

Current



Static Cataloging

SCION Transformation

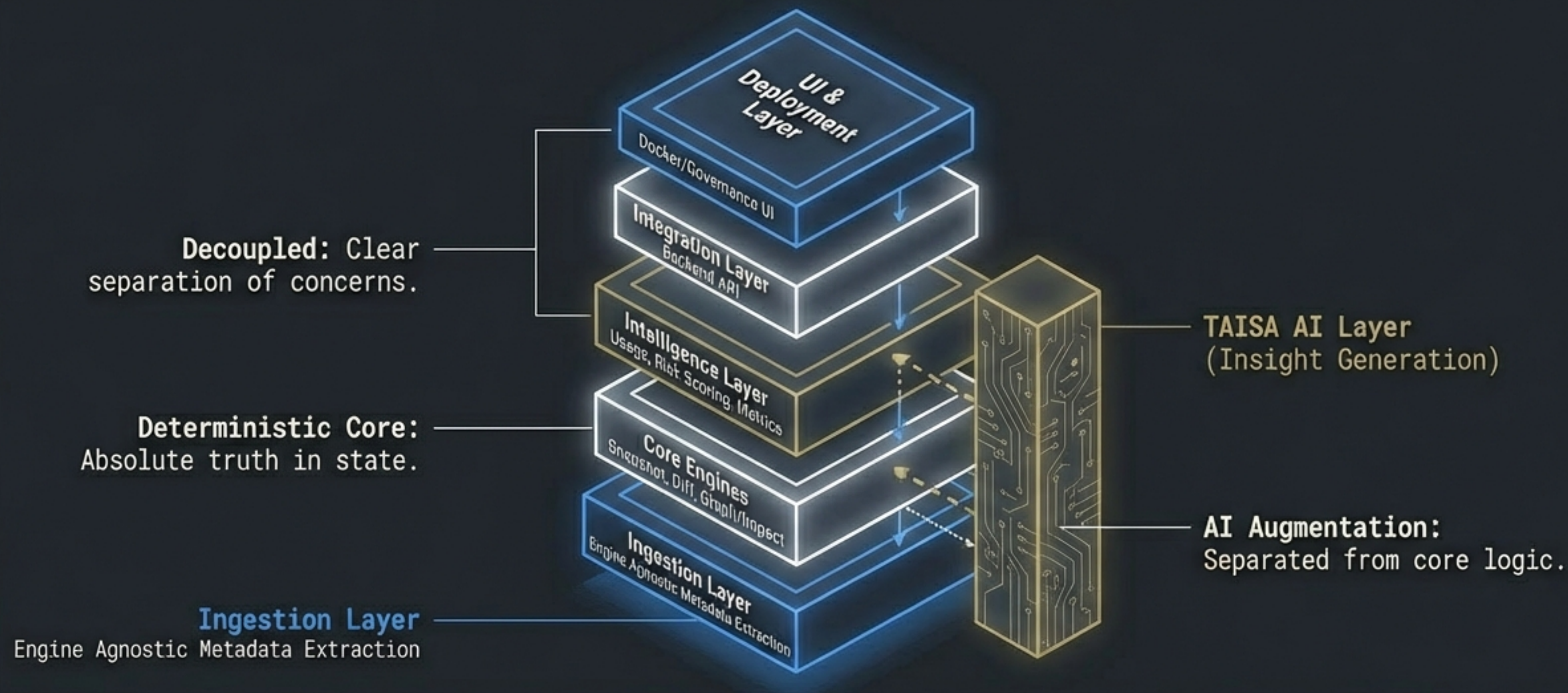
Future



Active Measurement & AI

- 🔧 **Definition:** An Intelligent Governance and Structural Intelligence Platform.
- 🧠 **AI Integration:** Native "Glass Box" AI (TAISA) vs. bolted-on features.
- 🎯 **Differentiation:** Focus on auditability, metrics, and deterministic accuracy.
- 🔗 **End State:** A system that doesn't just store metadata, but explains it.

High-Level Layered Architecture



Detailed Module Breakdown & Complexity

Module Name	Description	Est. Complexity
Metadata Ingestion	Template-based, multi-engine support	[Medium]
Snapshot Engine	Versioned history, transactional consistency	[Low]
Diff Engine	Deterministic change detection	[High]
Graph & Impact	Recursive dependency/lineage	[High]
Usage Engine	Log ingestion & frequency scoring	[Medium]
Metrics & TAISA	Vulnerability indexing & AI interpretation	[Medium]
UI/API/Deploy	Dockerized runtime & visualization	[Medium]

Phase 1: DNA-Ready Scope (MVP)

- 
-  **Ingestion:** Hardened connectors for Teradata/Postgres.
 -  **Core Logic:** Versioned snapshot tracking & deterministic diffs.
 -  **Impact:** Recursive analysis with usage-weighted risk scoring.
 -  **Intelligence:** Structural metrics and TAISA integration.
 -  **Access:** Secure API and Governance Intelligence UI.
 -  **Infrastructure:** Fully dockerized deployment.
- 

Timeline Target: 22–26 Weeks (5–6 Months)

Required Team Profile: Internal Development Model



TECHNICAL LEAD

Load: 1 FTE

Skills:

Architecture, SQL,
Modular Systems,
Backend Logic.

Note: Owns the Glass
Box logic.



BACKEND ENGINEER

Load: 1 FTE

Skills:

Python/FastAPI,
Testing, API
Orchestration.

Note: Executes the
build.



UI / VIZ SUPPORT

Load: 0.5 FTE (Shared)

Skills:

Dashboarding, Graph
Visualization.

Note: Optional /
Shared Resource.

100% Internal Initiative. Knowledge is shared and documented, not siloed.

Delivery Timeline by Team Structure

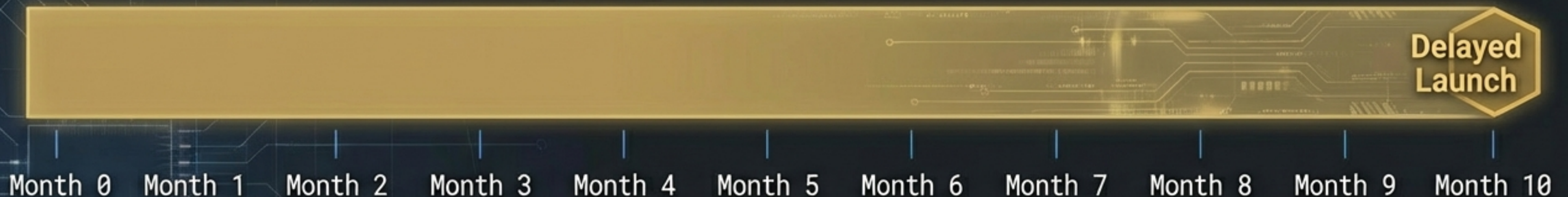
Month 0 Month 1 Month 2 Month 3 Month 4 Month 5 Month 6 Month 7 Month 8 Month 9 Month 10

Scenario A (Recommended): Lead + Backend Engineer



Parallel 'Shadow Mode' runs during development to validate logic and reduce risk.

Scenario B (Constrained): Single Engineer



SCION vs. Kalido: The Structural Advantage

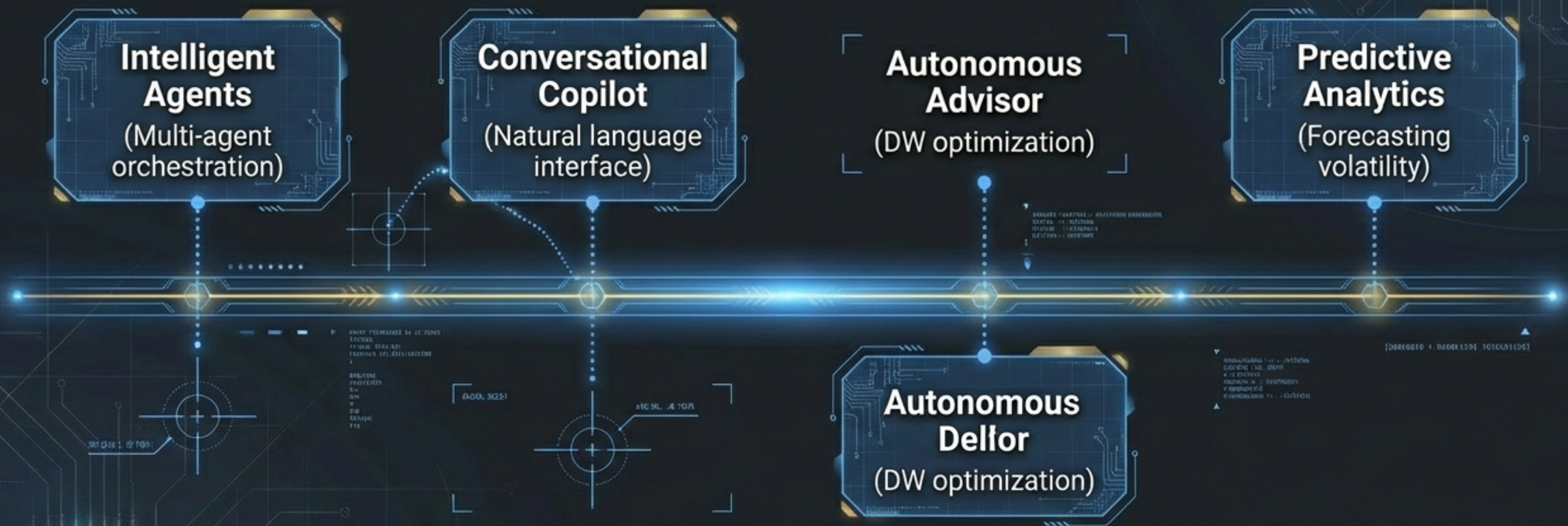
KALIDO

- ✗ Black-box logic (Non-auditable)
- ✗ High recurring costs (~\$480K/yr)
- ✗ Opaque scoring mechanisms
- ✗ Limited extensibility

SCION

- ✓ Glass-Box fully auditable architecture
- ✓ Zero licensing fees; Full IP ownership
- ✓ Native TAISA integration
- ✓ Quantified structural metrics (Volatility, Risk)

Phase 2: Strategic Expansion (Post DNA-Ready)



These capabilities are enabled by the architecture but are NOT in Phase 1 scope.

Controlled Rollout Plan



Objective: Zero disruption to operations.

Risk Analysis & Mitigation Strategies

RISK CHALLENGE	ENGINEERING MITIGATION
Scope Creep	Strict Phase 1 boundaries (DNA-Ready only).
SQL Parsing Complexity	Modular design and incremental engine support.
Performance Scaling	Containerized deployment and scalable architecture.
DNA Alignment	Native design; built FOR DNA, not adapted to it.
Adoption	'Shadow Mode' validation to prove accuracy.

Executive Recommendation

- ✓ **Financially Justified:** Immediate ROI via license elimination (~\$480K/yr).
- ✓ **Technically Viable:** Proven MVP & clear modular architecture.
- ✓ **Internally Executable:** Fits team capabilities; no external consultants.
- ✓ **Controlled Risk:** Phased rollout ensures no disruption.

> REQUEST: APPROVAL TO INITIATE
PHASE 1 DEVELOPMENT _